

DATA VALIDATION SYSTEM

JAVA SWING APPLICATION

WRITTEN SYSTEM DOCUMENTATION

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IN MICROSOFT WORD, RIGHT-CLICK THIS TABLE AND SELECT “UPDATE FIELD” TO GENERATE/UPDATE PAGE NUMBERS AUTOMATICALLY.

1. INTRODUCTION

THE DATA VALIDATION SYSTEM IS A JAVA SWING GRAPHIC USER INTERFACE DEVELOPED TO CAPTURE AND VALIDATE BASIC PERSONAL/STUDENT INFORMATION. THE FORM CAPTURES A USER'S FULL NAME, ID OR PASSPORT, AGE, GENDER, CONTACT NUMBER AND EMAIL ADDRESS.

THE MAIN PURPOSE OF THE SYSTEM IS TO PREVENT INCORRECT OR INCOMPLETE INFORMATION FROM BEING ACCEPTED. THE APPLICATION USES CONDITIONAL STATEMENTS (IF, ELSE IF AND ELSE), REGULAR EXPRESSIONS, STRING LENGTH CHECKS AND DATE VALIDATION TO PROVIDE FEEDBACK TO THE USER. VALIDATION FEEDBACK IS DISPLAYED NEXT TO THE RELEVANT FIELD.

2. SYSTEM DESIGN

BEFORE CODING, THE PLANNED DESIGN WAS BASED ON A SIMPLE FORM LAYOUT THAT GROUPS RELATED INFORMATION TOGETHER. THE INTERFACE USES LABELS BESIDE TEXT FIELDS SO THAT THE USER CAN CLEARLY IDENTIFY WHAT MUST BE ENTERED. GENDER IS PROVIDED USING RADIO BUTTONS, WHILE SAVE, CLEAR AND EXIT BUTTONS ARE PLACED AT THE BOTTOM OF THE FORM.

THE DESIGN WAS INTENDED TO BE EASY TO UNDERSTAND AND USE. A STRAIGHTFORWARD FORM REDUCES UNNECESSARY NAVIGATION AND ALLOWS USERS TO SEE VALIDATION FEEDBACK CLOSE TO THE FIELD THEY ARE COMPLETING. THE APPLICATION WAS CREATED USING JAVA SWING COMPONENTS IN NETBEANS.

SCREENSHOT TO INSERT

T3306 VALIDx

NAME:

CONTACT NO:

STUDENT ID:

AGE:

GENDER:

☐ **MALE**

☐ **FEMALE**

EMAIL:

SAVE CLEAR HELP EXIT

FIGURE 1: DATA VALIDATION SYSTEM INTERFACE

3. FIELDS

FIELD	COMPONENT	PURPOSE
FULL NAME	JTEXTFIELD	CAPTURES THE USER'S FULL NAME.
ID/STUDENT NUMBER	JTEXTFIELD	CAPTURES A 13-DIGIT IDENTIFICATION/STUDENT NUMBER.
AGE	JTEXTFIELD	CAPTURE THE USER'S AGE.
GENDER	JRADIOBUTTON	ALLOWS THE USER TO SELECT MALE OR FEMALE.
CONTACT NUMBER	JTEXTFIELD	CAPTURES A 10-DIGIT CONTACT NUMBER.
EMAIL ADDRESS	JTEXTFIELD	CAPTURES THE USER'S EMAIL ADDRESS.

4. VALIDATION CHECKS

THE SYSTEM VALIDATES INPUT WHEN THE USER PRESSES ENTER IN A TEXT FIELD OR SELECTS SAVE. THE VALIDATION METHODS USE IF/ELSE IF/ELSE STATEMENTS SO THAT THE INPUT IS CHECKED IN A LOGICAL ORDER.

FIELD	PRESENCE	TYPE	LENGTH	FORMAT	RANGE/RULE
FULL NAME	YES	LETTERS/SPACES	2–30 CHARACTER S	ALPHABETIC TEXT	LENGTH MUST BE WITHIN LIMITS
ID/STUDENT NUMBER	YES	DIGITS ONLY	EXACTLY 13 DIGITS	NUMERIC	EXACTLY 13 DIGITS
AGE	YES	INT	0-120	NUMERIC	MUST BE BETWEEN 0 AND 120
GENDER	YES	SELECTION	N/A	MALE/FEMALE OPTION	ONE OPTION MUST BE SELECTED
CONTACT NUMBER	YES	DIGITS ONLY	EXACTLY 10 DIGITS	NUMERIC	EXACTLY 10 DIGITS
EMAIL ADDRESS	YES	TEXT	MAXIMUM 100 CHARACTER S	EMAIL PATTERN	MUST CONTAIN A VALID EMAIL STRUCTURE

4.1 NAME VALIDATION

THE NAME FIELD FIRST CHECKS WHETHER IT IS EMPTY. IT THEN CHECKS THAT ONLY LETTERS AND SPACES ARE USED. FINALLY, IT CHECKS THAT THE NAME IS BETWEEN 2 AND 30 CHARACTERS. THIS PREVENTS BLANK, NUMERIC OR EXCESSIVELY LONG NAMES FROM BEING ACCEPTED.

```
STRING NAME = TXTNAME.GETTEXT();

IF (NAME.ISEMPTY()) {

    LBLNAME.SETTEXT("FIELD IS EMPTY");

    LBLNAME.SETFOREGROUND(COLOR.RED);
}

ELSE IF(NAME.MATCHES("[A-Z A-Z]+")){

    LBLNAME.SETTEXT("VALIDATED");

    LBLNAME.SETFOREGROUND(COLOR.GREEN);

}

ELSE IF(NAME.LENGTH() < 2 || NAME.LENGTH() > 30){

    LBLNAME.SETTEXT("ENTER FULL NAMES");

    LBLNAME.SETFOREGROUND(COLOR.RED);

}
```

4.2 ID/STUDENT NUMBER VALIDATION

THE ID FIELD CHECKS FOR AN EMPTY VALUE, CHECKS THAT ONLY DIGITS ARE ENTERED, AND THEN CHECKS THAT THE VALUE CONTAINS EXACTLY 13 DIGITS.

```
STRING ID = TXTSTUDENT.GETTEXT();

IF (STUDENTID.ISEMPTY()){

    LBLSTUDENT.SETTEXT("FIELD IS EMPTY");

    LBLSTUDENT.SETFOREGROUND(COLOR.RED);

}

ELSE IF(!STUDENTID.LENGTH().MATCHES("\\d+")){

    LBLSTUDENT.SETTEXT("INVALID INPUT");

    LBLSTUDENT.SETFOREGROUND(COLOR.RED);

}

ELSE IF(STUDENTID.LENGTH() != 13){

    LBLSTUDENT.SETTEXT("13 DIGITS ONLY");

    LBLSTUDENT.SETFOREGROUND(COLOR.RED);

}

ELSE{

    LBLSTUDENT.SETTEXT("VALIDATED");

    LBLSTUDENT.SETFOREGROUND(COLOR.GREEN);

}

}
```

4.3 DATE OF BIRTH VALIDATION

THE AGE FIRST CHECKS WHETHER THE FIELD IS EMPTY. IT THEN CHECKS WHETHER THE ENTERED VALUE IS AN INTEGER. THE SYSTEM CHECKS THAT THE AGE IS BETWEEN 0 AND 120.

```
INT AGE = INTEGER.PARSEINT(TXTAGE.GETTEXT());
    IF(AGE < 0 || AGE > 120){

        LBLAGE.SETTEXT("BETWEEN 0 AND 120");

        LBLAGE.SETFOREGROUND(COLOR.RED);

    ELSE{

        LBLAGE.SETTEXT("VALIDATED");

        LBLAGE.SETFOREGROUND(COLOR.GREEN);

    }

}
```

```
}
```

4.4 GENDER VALIDATION

THE GENDER FIELD USES TWO RADIO BUTTONS. SELECTING ONE OPTION DESELECTS THE OTHER. WHEN SAVE IS PRESSED, THE SYSTEM CHECKS THAT AT LEAST ONE GENDER OPTION HAS BEEN SELECTED.

```
PRIVATE VOID (GENDERVALID){  
    IF (MALEBTN.ISSELECTED){  
        LBLGENDER.SETTEXT("MALE!");  
        LBLGENDER.SETFOREGROUND(COLOR.GREEN);  
    }  
    ELSE IF (FEMALEBTN.ISSELECTED){  
        LBLGENDER.SETTEXT("FEMALE!");  
        LBLGENDER.SETFOREGROUND(COLOR.PINK);  
    }  
}
```

4.5 CONTACT NUMBER VALIDATION

THE CONTACT NUMBER MUST NOT BE EMPTY, MUST CONTAIN DIGITS ONLY, AND MUST CONTAIN EXACTLY 10 DIGITS.

```
STRING CONTACTNo = TXTCONTACT.GETTEXT();  
  
IF (CONTACT.ISEMPTY()) {  
    LBLCONTACT.SETTEXT("FIELD IS EMPTY");  
    LBLCONTACTNo.SETFOREGROUND(COLOR.RED);  
}  
  
ELSE IF(!CONTACTNo.MATCHES("\\d+")){  
    LBLCONTACT.SETTEXT("ENTER 10 DIGITS ONLY!");  
    LBLCONTACTNo.SETFOREGROUND(COLOR.RED);  
}  
  
ELSE{  
    LBLCONTACT.SETTEXT("VALIDATED!");  
    LBLCONTACTNo.SETFOREGROUND(COLOR.GREEN);  
}
```

4.6 EMAIL VALIDATION

THE EMAIL FIELD CHECKS THAT A VALUE IS PRESENT, USES A BASIC EMAIL PATTERN, AND DOES NOT EXCEED 100 CHARACTERS. THE REGULAR EXPRESSION IS USED AS THE FORMAT CHECK.

```
STRING EMAIL = TXTEMAIL.GETTEXT();

IF (EMAIL.ISEMPTY()) {
    LBLEMAIL.SETTEXT("FIELD IS EMPTY");

    LBLEMAIL.SETFOREGROUND(COLOR.RED);
} ELSE IF (!EMAIL.CONTAINS("@GMAIL.COM")){

    LBLEMAIL.SETTEXT("VALIDATE!");

    LBLEMAIL.SETFOREGROUND(COLOR.RED);

}

ELSE{

    LBLEMAIL.SETTEXT("INVALID INPUT");

    LBLEMAIL.SETFOREGROUND(COLOR.RED);

}

}
```

5. TESTING

TESTING WAS DESIGNED TO DEMONSTRATE THAT INVALID INPUT IS REJECTED AND THAT VALID INPUT IS ACCEPTED. THE EVIDENCE SHOULD SHOW THE VALUE ENTERED AND THE FEEDBACK PRODUCED BY THE APPLICATION. SCREENSHOTS SHOULD BE TAKEN DIRECTLY FROM THE RUNNING APPLICATION.

5.1 BEFORE TESTING: INVALID INPUT

THE FOLLOWING SCREENSHOTS MUST BE CAPTURED WHILE INVALID VALUES ARE ENTERED. EACH SCREENSHOT SHOULD SHOW THE CORRESPONDING ERROR MESSAGE.

SCREENSHOT TO INSERT

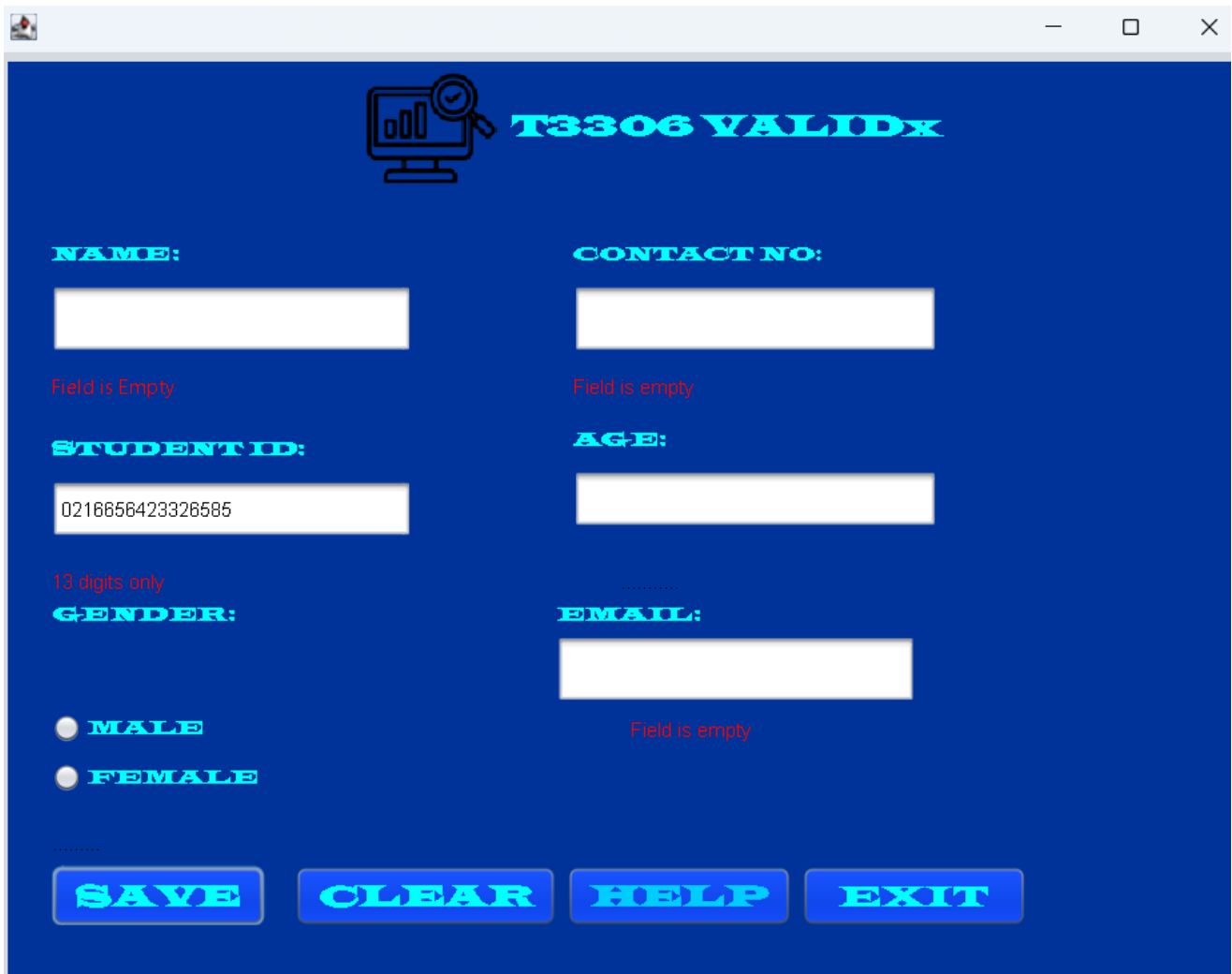
The screenshot shows a web application window titled "T3306 VALIDx" with a blue background. At the top left is an icon of a computer monitor with a bar chart and a magnifying glass. The form contains the following fields and labels:

- NAME:** A text input field containing "thato108". Below it, the text "Letters Only" is displayed in red.
- CONTACT NO:** An empty text input field. Below it, the text "Field is empty" is displayed in red.
- STUDENTID:** An empty text input field. Below it, the text "Field is empty" is displayed in red.
- AGE:** An empty text input field.
- GENDER:** Two radio button options: **MALE** and **FEMALE**.
- EMAIL:** An empty text input field. Below it, the text "Field is empty" is displayed in red.

At the bottom of the form are four buttons: **SAVE**, **CLEAR**, **HELP**, and **EXIT**.

FIGURE 2: INVALID FULL NAME

SCREENSHOT TO INSERT



The screenshot shows a software application window titled "T3306 VALIDx". The window has a dark blue background and a light blue title bar with standard Windows window controls (minimize, maximize, close). At the top center, there is a logo consisting of a computer monitor displaying a bar chart, with a magnifying glass icon over it. Below the logo, the text "T3306 VALIDx" is displayed in a large, bold, white font.

The main area of the window contains several input fields and labels, all in white text on the dark blue background:

- NAME:** A text input field. Below it, the text "Field is Empty" is displayed in red.
- CONTACT NO:** A text input field. Below it, the text "Field is empty" is displayed in red.
- STUDENT ID:** A text input field containing the value "0216656423326585". Below it, the text "13 digits only" is displayed in red.
- AGE:** A text input field.
- GENDER:** Two radio button options: **MALE** and **FEMALE**. Both are currently unselected.
- EMAIL:** A text input field. Below it, the text "Field is empty" is displayed in red.

At the bottom of the window, there are four buttons: **SAVE**, **CLEAR**, **HELP**, and **EXIT**. Each button has a blue gradient background and white text.

FIGURE 3: INVALID ID/STUDENT NUMBER

SCREENSHOT TO INSERT

 **T3306 VALIDx**

NAME:

Field is Empty

CONTACT NO:

Field is empty

STUDENT ID:

Field is empty

AGE:

Between 0 and 120

GENDER:
☐ **MALE**
☐ **FEMALE**
Invalid selection

EMAIL:

Field is empty

SAVE **CLEAR** **HELP** **EXIT**

FIGURE 4: INVALID AGE

SCREENSHOT TO INSERT

 **T3306 VALIDx**

NAME:

Field is Empty

CONTACT NO:

Field is empty

STUDENT ID:

Field is empty

AGE:

GENDER:
☐ **MALE**
☐ **FEMALE**

EMAIL:

Field is empty

.....

SAVE **CLEAR** **HELP** **EXIT**

FIGURE 5: NO GENDER SELECTED

SCREENSHOT TO INSERT

 **T3306 VALIDx**

NAME:

Field is Empty

CONTACT NO:

Enter 10 digits only!

STUDENT ID:

Field is empty

AGE:

GENDER:
☐ **MALE**
☐ **FEMALE**

EMAIL:

Field is empty

.....

SAVE **CLEAR** **HELP** **EXIT**

FIGURE 6: INVALID CONTACT NUMBER

SCREENSHOT TO INSERT

The screenshot shows a web application window titled "T3306 VALIDx". The interface has a dark blue background with white text and input fields. At the top left is a logo of a computer monitor with a bar chart and a checkmark. The form contains the following elements:

- NAME:** A white text input field with the feedback "Field is Empty" below it.
- CONTACT NO:** A white text input field with the feedback "Field is empty" below it.
- STUDENT ID:** A white text input field with the feedback "Field is empty" below it.
- AGE:** A white text input field.
- GENDER:** Two radio button options: **MALE** and **FEMALE**.
- EMAIL:** A white text input field containing "BJKSDCKDK" with the feedback "INVALID INPUT" below it.
- Buttons:** Four blue buttons with white text: **SAVE**, **CLEAR**, **HELP**, and **EXIT**.

FIGURE 7: INVALID EMAIL ADDRESS

5.2 AFTER TESTING: VALID INPUT

REPEAT THE SAME TESTS USING VALID VALUES. THE FEEDBACK SHOULD CHANGE TO "VALID" FOR THE CORRESPONDING FIELDS. THE SCREENSHOTS BELOW SHOULD SHOW THE ACCEPTED VALUES AND VALIDATION FEEDBACK.

SCREENSHOT TO INSERT

T3306 VALIDx

NAME:
Thato Chapatso
Validated

CONTACT NO:
Field is empty

STUDENT ID:
Field is empty

AGE:
Field is empty

GENDER:
☐ MALE
☐ FEMALE

EMAIL:
Field is empty

SAVE CLEAR HELP EXIT

FIGURE 8: VALID FULL NAME

SCREENSHOT TO INSERT

T3306 VALIDx

NAME:

CONTACT NO:

STUDENT ID:

AGE:

GENDER:

EMAIL:

MALE

FEMALE

SAVE **CLEAR** **HELP** **EXIT**

FIGURE 9: VALID ID/STUDENT NUMBER

SCREENSHOT TO INSERT

T3306 VALIDx

NAME:

Field is Empty

CONTACT NO:

Field is empty

STUDENT ID:

Field is empty

AGE:

Validated!

GENDER:
☐ **MALE**
☐ **FEMALE**
NO GENDER S...

EMAIL:

Field is empty

SAVE **CLEAR** **HELP** **EXIT**

FIGURE 10: VALID AGE

SCREENSHOT TO INSERT

 **T3306 VALIDx**

NAME:

Field is Empty

CONTACT NO:

Field is empty

STUDENT ID:

Field is empty

AGE:

GENDER:
☒ **MALE**
☐ **FEMALE**

EMAIL:

Field is empty

SAVE CLEAR HELP EXIT

FIGURE 11: VALID GENDER SELECTION

SCREENSHOT TO INSERT

The screenshot shows a web application window titled "T3306 VALIDx". The interface has a dark blue background with white text and form elements. At the top left, there is an icon of a computer monitor with a bar chart and a magnifying glass. The form contains the following fields and labels:

- NAME:** A text input field that is currently empty. Below it, the text "Field is Empty" is displayed in red.
- CONTACT NO:** A text input field containing the number "0898569416". Below it, the text "Validated" is displayed in green.
- STUDENT ID:** A text input field that is currently empty. Below it, the text "Field is empty" is displayed in red.
- AGE:** A text input field that is currently empty.
- GENDER:** Two radio button options: **MALE** and **FEMALE**. The **FEMALE** option is selected.
- EMAIL:** A text input field that is currently empty. Below it, the text "Field is empty" is displayed in red.

At the bottom of the form, there are four buttons: **SAVE**, **CLEAR**, **HELP**, and **EXIT**.

FIGURE 12: VALID CONTACT NUMBER

SCREENSHOT TO INSERT

The screenshot shows a web application window titled "T3306 VALIDx". The interface has a dark blue background with white text and form elements. At the top left, there is an icon of a computer monitor with a bar chart and a magnifying glass. The form consists of several input fields and a radio button group. The "NAME:" field is empty, with a red "Field is Empty" message below it. The "CONTACT NO:" field is also empty, with a red "Field is empty" message below it. The "STUDENT ID:" field is empty, with a red "Field is empty" message below it. The "AGE:" field is empty. The "GENDER:" section has two radio buttons: "MALE" (selected) and "FEMALE". The "EMAIL:" field contains the text "thatochapatso@gmail.com" and has a green "validated!" message below it. At the bottom, there are four buttons: "SAVE", "CLEAR", "HELP", and "EXIT".

FIGURE 13: VALID EMAIL ADDRESS

5.3 ALL INPUT IS VALID

THE FINAL TEST SHOULD CONTAIN VALID INFORMATION IN EVERY FIELD AND A SELECTED GENDER. PRESS SAVE AND CAPTURE THE COMPLETE FORM. THIS SCREENSHOT IS IMPORTANT EVIDENCE THAT ALL FIELDS CAN PASS VALIDATION TOGETHER.

SCREENSHOT TO INSERT

T3306 VALIDx

NAME:
Thato Chapatso
Validated

CONTACT NO:
0898569416
Validated

STUDENT ID:
0210065440088
Validated

AGE:
24
Validated!

GENDER:
☒ **MALE**
☐ **FEMALE**
 Male
 Validated!

EMAIL:
thatochapatso@gmail.com
validated!

SAVE CLEAR HELP EXIT

FIGURE 14: COMPLETED FORM WITH ALL INPUT VALID

6. CODE

THE APPLICATION WAS CREATED IN JAVA SWING. THE KEY SECTIONS OF THE SOURCE CODE ARE SHOWN BELOW RATHER THAN REPRODUCING THE ENTIRE GENERATED NETBEANS FILE. THE CODE DEMONSTRATES THE VALIDATION LOGIC, EVENT HANDLING AND BUTTON FUNCTIONALITY.

6.1 VALIDATION FEEDBACK METHODS

```
PRIVATE VOID VALIDATED(JLABEL LABEL, STRING MESSAGE) {
    LABEL.SETTEXT(MESSAGE);
    LABEL.SETFOREGROUND(COLOR.GREEN);
}
```

```
PRIVATE VOID NOTVALID(JLABEL LABEL, STRING MESSAGE) {
    LABEL.SETTEXT(MESSAGE);
```

```

    LABEL.SETFOREGROUND(COLOR.RED);
}

```

THESE TWO METHODS CENTRALISE THE FEEDBACK USED BY THE VALIDATION SYSTEM. THE LABEL RECEIVES A MESSAGE DEPENDING ON WHETHER THE INPUT PASSES OR FAILS VALIDATION.

6.2 SAVE BUTTON

```

PRIVATE VOID SAVEBTNActionPerformed(JAVA.AWT.EVENT.ACTIONEVENT EVT) {
    NAMETXTActionPerformed(NULL);
    IDTXTActionPerformed(NULL);
    BIRTHTXTActionPerformed(NULL);
    NUMBERTXTActionPerformed(NULL);
    EMAILTXTActionPerformed(NULL);

    IF (!JRadioButton1.isSelected() && !JRadioButton2.isSelected()) {
        NOTVALID(GENDERVAL, "SELECT GENDER");
    } ELSE {
        VALIDATED(GENDERVAL, "VALID");
    }
}

```

THE SAVE BUTTON RUNS THE VALIDATION METHOD FOR EACH TEXT FIELD AND THEN CHECKS WHETHER A GENDER OPTION HAS BEEN SELECTED. THIS ALLOWS THE WHOLE FORM TO BE CHECKED FROM ONE BUTTON.

6.3 CLEAR AND EXIT BUTTONS

```

PRIVATE VOID EXITBTNActionPerformed(JAVA.AWT.EVENT.ACTIONEVENT EVT) {
    DISPOSE();
}

PRIVATE VOID CLEARBTNActionPerformed(JAVA.AWT.EVENT.ACTIONEVENT EVT) {
    TXTNAME.setText("");
    TXTSTUDENT.setText("");
    TXTCONTACT.setText("");
    TXTAGE.setText("");
    TXTEMAIL.setText("");

    MALEBTN.isSelected(false);
    FEMALEBTN.setSelected(false);

    LBLNAME.setText(" ");

    LBLSTUDENT.setText(" ");

    LBLCONTACT.setText(" ");

    LBLAGE.setText(" ");

    LBLEMAIL.setText(" ");

    LBLGENDER.setText(" ");
}

```

EXIT CLOSES THE APPLICATION WINDOW. CLEAR REMOVES THE ENTERED VALUES AND RESETS THE GENDER SELECTIONS. THE COMPLETE SOURCE ALSO CLEARS THE VALIDATION LABELS.

7. GITHUB REPOSITORY

GITHUB REPOSITORY: [INSERT FULL CLICKABLE GITHUB REPOSITORY LINK HERE]

A MARKER CAN CLONE THE PROJECT BY OPENING THE REPOSITORY LINK, SELECTING THE CODE OPTION, COPYING THE REPOSITORY URL, AND CLONING IT IN NETBEANS OR ANOTHER JAVA IDE. THE PROJECT SHOULD THEN BE OPENED AS A NETBEANS JAVA PROJECT AND THE VALIDATION CLASS SHOULD BE RUN. THE MAIN METHOD CREATES THE JFRAME AND MAKES THE APPLICATION VISIBLE.

REPOSITORY SETUP NOTE: ENSURE THAT THE PACKAGE NAME AND PROJECT STRUCTURE REMAIN UNCHANGED WHEN CLONING THE PROJECT. IF THE PROJECT USES NETBEANS-GENERATED FILES, THE COMPLETE PROJECT FOLDER SHOULD BE COMMITTED TO GITHUB RATHER THAN ONLY THE JAVA SOURCE FILE.

8. CONCLUSION

THE DATA VALIDATION SYSTEM PROVIDED PRACTICAL EXPERIENCE IN DESIGNING AND IMPLEMENTING A JAVA SWING FORM. ONE OF THE MAIN CHALLENGES WAS UNDERSTANDING HOW EVENT METHODS AND IF/ELSE STATEMENTS CONTROL VALIDATION. THE PROJECT DEMONSTRATED THAT INPUT SHOULD BE CHECKED IN STAGES: FIRST CHECKING WHETHER A VALUE EXISTS, THEN CHECKING ITS TYPE OR FORMAT, FOLLOWED BY LENGTH AND OTHER RULES.

ANOTHER IMPORTANT LESSON WAS LEARNING HOW REGULAR EXPRESSIONS CAN BE USED TO VALIDATE NAMES, NUMBERS AND EMAIL ADDRESSES, WHILE AGE CAN BE USED TO DETERMINE IF THE USER CAN DO THE INSTRUCTION. THE PROJECT ALSO SHOWED THE IMPORTANCE OF TESTING BOTH INVALID AND VALID INPUT RATHER THAN ASSUMING THAT THE APPLICATION WORKS AFTER THE CODE COMPILES.

OVERALL, THE PROJECT IMPROVED MY UNDERSTANDING OF JAVA SWING COMPONENTS, EVENT HANDLING, CONDITIONAL STATEMENTS, VALIDATION RULES AND USER FEEDBACK. THE TESTING SCREENSHOTS PROVIDE EVIDENCE THAT THE SYSTEM RESPONDS TO DIFFERENT TYPES OF INPUT.

9. REFERENCES

THE FOLLOWING REFERENCES ARE SUITABLE FOR THE TECHNOLOGIES USED IN THE PROJECT. ADD ANY ADDITIONAL TUTORIALS OR LEARNING MATERIALS ACTUALLY USED DURING DEVELOPMENT.

- ORACLE. JAVA DOCUMENTATION. [HTTPS://DOCS.ORACLE.COM/EN/JAVA/](https://docs.oracle.com/en/java/)
- ORACLE. JAVA SWING DOCUMENTATION. [HTTPS://DOCS.ORACLE.COM/JAVASE/TUTORIAL/UISWING/](https://docs.oracle.com/javase/tutorial/uiswing/)

- ORACLE. LOCALDATE API DOCUMENTATION.
[HTTPS://DOCS.ORACLE.COM/EN/JAVA/JAVASE/](https://docs.oracle.com/en/java/javase/)
- NETBEANS IDE DOCUMENTATION. [HTTPS://NETBEANS.APACHE.ORG/](https://netbeans.apache.org/)

10. SUBMISSION CHECKLIST

- ☐ COVER PAGE COMPLETED WITH STUDENT NUMBER, COLLEGE NAME, LECTURER AND COLLEGE LOGO.
- ☐ TABLE OF CONTENTS UPDATED AUTOMATICALLY IN MICROSOFT WORD.
- ☐ SYSTEM INTERFACE SCREENSHOT INSERTED.
- ☐ INVALID-INPUT SCREENSHOTS INSERTED FOR EVERY FIELD.
- ☐ VALID-INPUT SCREENSHOTS INSERTED FOR EVERY FIELD.
- ☐ COMPLETED-FORM SCREENSHOT INSERTED.
- ☐ GITHUB REPOSITORY LINK INSERTED AND TESTED.
- ☐ ALL SCREENSHOTS HAVE FIGURE NUMBERS/CAPTIONS.
- ☐ DOCUMENT PROOFREAD FOR SPELLING AND GRAMMAR.
- ☐ FINAL DOCUMENT SAVED/SUBMITTED AS ONE DOCX OR EXPORTED TO PDF.